
Merging Specialist Diffusion Models for Multi-Task Astronomical Image Restoration

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Abstract

Astronomical image-processing pipelines typically chain several independent restoration steps, each best solved by a dedicated tool. We investigate whether such a pipeline can be collapsed into a *single* set of weights via post-hoc model merging, avoiding both the cost of serving multiple networks and the need for joint multi-task training data. In particular, we focus on *star removal*, *defect correction* and *super-resolution*. Three *v-prediction* DDPMs are independently fine-tuned (one per task) from a common, purpose-pretrained initialization θ_0 , and their task vectors are combined with Task Arithmetic, TIES-Merging, and DARE-TIES. Crucially, because initial zero-shot merging configurations failed to meet performance expectations due to destructive weight interference, we introduce a brief post-merge fine-tuning stage for all three methods. Merging and fine-tuning hyper-parameters are chosen via a *two-stage grid search* evaluated on all 7 combinations of the three degradations, using manually-curated Hubble Legacy Archive (HLA) imagery. Code is available [here](#).

1. Introduction

Restoring astronomical images often requires multiple sequential corrections, including star removal, detector defect restoration (dead/hot pixels), and super-resolution. A common deep learning approach is to train one specialist model per task, but this requires storing and executing multiple networks.

Model merging combines independently fine-tuned checkpoints into a single model without joint training. While ef-

fective for classification and language models, we find that it performs poorly when applied zero-shot to conditional diffusion models for low-level image restoration.

Our experiments indicate that zero-shot merging of diffusion specialists suffers from destructive interference that is not always reflected by pixel-wise metrics but becomes evident under perceptual evaluation. We further show that a short, balanced post-merge fine-tuning stage consistently improves the performance of Task Arithmetic, TIES, and DARE-TIES, suggesting that post-merge adaptation has a greater impact than the specific merging algorithm.

2. Related Work

Diffusion models. Denoising Diffusion Probabilistic Models (DDPMs) learn to invert a fixed noising process (Ho et al., 2020). We adopt the *v-prediction* parameterization (Salimans & Ho, 2022), a cosine noise schedule (Nichol & Dhariwal, 2021), and classifier-free-style condition dropout so that the network retains some unconditional generative capacity; the backbone is a conditional U-Net (Ronneberger et al., 2015) with self-attention (Vaswani et al., 2017) at low resolutions.

Model merging. Task Arithmetic (Ilharco et al., 2023) combines task vectors by adding scaled weight differences to a shared pretrained model. TIES-Merging (Yadav et al., 2023) reduces interference by trimming low-magnitude updates and resolving sign conflicts, while DARE (Yu et al., 2024) further sparsifies task vectors through random pruning and rescaling. All methods require the same pretrained origin, motivating our shared- θ_0 stage (Sec. 3.2).

These techniques were developed and are still predominantly evaluated on image classifiers and language models. Their use in diffusion has so far centred on either coarse weight averaging of models trained on different data shards, or on composing lightweight LoRA adapters for concept- and style-level personalization (e.g. concept sliders), where each adapter edits a semantic direction rather than solving a different restoration problem. To our knowledge, coordinate-level merging (Task Arithmetic, TIES,

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DARE-TIES) of *fully* fine-tuned diffusion specialists that each solve a distinct low-level restoration task – and the observation that such merges require a post-merge repair stage to be usable – has not been studied; this is the setting we address. **[TODO: add references to back up these claims, and search on the web if this is true]**

3. Method

3.1. Dataset

Source imagery and pre-processing. We start from a manually curated set of color, already-drizzled mosaics from the HLA (Space Telescope Science Institute (STScI), 2006). Raw frames are converted to a working dynamic range via a nonlinear stretch before any degradation is applied. Applying a uniform, global stretching function across the entire dataset yielded highly suboptimal results due to the vast diversity in exposure levels, noise floors, and dynamic ranges among different astronomical targets. To overcome this, an image-specific adaptive arcsinh stretch is performed on each pair before training. **[TODO: make this shorter]**

Dataset generation. For all three tasks the supervision target is always the fully clean, stretched image; only the *condition* (network input) differs. Each specialist therefore sees, as condition, an image degraded with only its own corruption, but is only ever supervised to recover the clean image – as opposed to θ_0 ’s pretext task (Sec. 3.2), where the degradation is deliberately *not* tied to the task the source image came from.

1. **Star removal.** StarNetV2 (Misiura, 2020) is used only to *isolate* real stars (by subtraction from the clean image), not as ground truth. The extracted star patches are re-injected onto pristine starless backgrounds via linear blending, yielding controlled input–target pairs.
2. **Dead/hot-pixel restoration.** Defective pixels are flagged by a robust detector on the residual from a 5×5 median filter: a pixel is defective when its channel residual exceeds 6.0σ , with σ from the MAD scaled by 1.4826. As HLA composites combine independent filter exposures, defects are sparse and usually confined to one channel.
3. **Super-resolution.** Each high-resolution image is degraded with Gaussian blur ($\sigma = 1.5$) and $\times 4$ cubic downsampling, then upsampled back to full resolution via bicubic interpolation to give a spatially aligned conditioning input for the DDPM.

After each of these processes is done, randomized patch sampling and data augmentation extract sub-images exclusively from regions verified by a geometric validity mask.

Combined-degradation sets. To evaluate merged models under realistically compounded corruption, we build 7 sets by applying the corresponding combinations of single-task degradations *in sequence* to the same clean target.

3.2. Shared initialization θ_0

All three specialists are fine-tuned from a common checkpoint θ_0 , obtained by pretraining the same architecture from a random initialization on a *generic conditional-restoration* pretext task, rather than starting each specialist from an independent random init, as required by the original merging methods themselves.

More specifically, θ_0 is trained to reconstruct clean images from synthetically degraded inputs. Degradations consist of 1–3 randomly selected operations from {Gaussian/Poisson-like noise, Gaussian blur, bicubic downsample-upsample, and small local occlusions}, with independently sampled parameters and a 10% probability of a fully zeroed condition (classifier-free guidance dropout (Ho & Salimans, 2022)). This decorrelates image content from degradation type, preventing the pretrained model from learning task-specific biases.

3.3. Architecture

[TODO: collapse all of this section into one single and shorter paragraph]

The following architecture remains *strictly identical* across θ_0 pre-training and all downstream fine-tuning tasks. The model relies on a conditional U-Net architecture (*Improve-dUNet*) that processes a 6-channel tensor obtained by concatenating the noisy latent x_t and the spatial conditioning map. The network features an initial feature extraction block followed by 4 downsampling and 4 upsampling stages. Downsampling is performed via strided convolutions (4×4 , stride 2), which progressively double the channel width at each stage ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$).

Each architectural stage is built around residual *UNet-Blocks* comprising two convolutional layers (3×3), Group Normalization (4 groups), and SiLU activations. Timestep embedding is achieved using a Sinusoidal projection layer ($C = 64$) followed by a two-layer multi-layer perceptron with GELU activation to output a 256-dimensional time embedding vector, which is linearly projected and added to the intermediate features of every residual block.

Multi-head self-attention (2 heads with Layer Normalization) is applied at the deepest encoder stage, the bottleneck, and the first decoder stage. The decoder upsamples features using transposed convolutions with skip connections, and a final convolution projects the 64-channel output to a 3-channel v -prediction.

3.4. Specialist fine-tuning and merging

Each specialist reuses θ_0 's architecture and the v -prediction framework, incorporating a cosine noise schedule over $T = 200$ diffusion steps, an exponential moving average (EMA) with $\beta = 0.999$, and the AdamW optimizer. Fine-tuning is conducted exclusively on each specialist's respective (condition, target) pairs from Sec. 3.1. The optimization utilizes a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4} , regulated by a cosine annealing learning rate scheduler over a total of 100 epochs.

Two-stage grid search. [TODO: explain how to do the two stages grid search from the notebook i sent you]

4. Results

4.1. Choosing the search objective

We first developed and validated the search objective on Task Arithmetic, iterating through several evaluation metrics; the best-performing configuration was then applied unchanged to TIES and DARE-TIES on the final iteration to maximise the results.

Raw, pooled MSE. The earliest attempts scored candidate merges by MSE against ground truth, first at a single hand-picked λ point and then over an actual grid, split into a two-stage. However both consistently picked $\lambda_{SU} = 0$: because the super-resolution folder carries by far the largest absolute MSE, any pooled objective is minimised by *suppressing* that task rather than balancing it.

Normalised MSE. We proceeded to measure each task's error relative to its own single-task expert, alongside PSNR/SSIM for a fuller picture. This normalised objective was the first to select a genuine three-task compromise instead of switching a task off entirely. Combined-degradation folders have no single expert to normalise against, so they were instead scored relative to θ_0 — i.e. how much merging improves on doing nothing.

Perceptual Z-score. The final refinement addresses the mismatch between low MSE and visibly poor super-resolution output: on any folder involving super-resolution, LPIPS replaces MSE. To make heterogeneous per-folder scores comparable, each folder is standardised with a per-folder Z-score $z = (x - \mu)/\sigma$ computed across the candidate grid, and the resulting Z-scores are averaged to rank candidates.

However, when troubleshooting the winning Task-Arithmetic merge we found a striking disagreement between the two families of metrics. The merge attains the *lowest* MSE of the three yet by far the *worst* LPIPS: pixel

Merging method	Zero-shot	Post-merge FT
Task Arith.	+0.58	−0.49
TIES	+0.63	−0.51
DARE-TIES	+0.71	−0.44

Table 1. Mean perceptual Z-score (averaged over the seven evaluation folders; lower is better) before and after post-merge fine-tuning. The brief fine-tune moves every method from a positive (worse-than-average) to a negative (better-than-average) Z-score, and the three methods converge to a comparable level.

error was rewarding a smoother but perceptually poorer prediction, and the gap survives much longer sampling, ruling out a sampling artifact. Although the task vectors are nearly orthogonal (cosine similarity ≈ 0.08 – 0.12), which normally favours task arithmetic, τ_{SU} 's much larger norm means even a small contribution from the other tasks perturbs its direction enough to visibly degrade quality.

Findings. [TODO: start the sentence better], the evaluation metric mattered more than the search algorithm. Raw pooled MSE optimises total pixel error rather than balanced multi-task quality, and therefore repeatedly suppresses super-resolution; normalising against per-task baselines removes this pooling artifact and recovers a genuine three-task compromise. Only a perceptual objective, however, exposes the task-vector interference that pixel-wise error cannot see, so any merge of these experts should be validated perceptually before it is trusted.

4.2. Effect of post-merge fine-tuning

Applied zero-shot, all three merging methods produced visible artifacts and degraded reconstruction fidelity, confirming the destructive interference discussed above. We therefore fine-tuned each merged checkpoint with a short, deliberately conservative schedule that initializes *both* the network and its EMA copy from the merged weights and trains on a small, balanced mixture of all seven degradation combinations (SR, IR, SU and their pairings and triple). The diffusion formulation, optimizer and schedule are kept identical to the specialists (Sec. 3.4), so that only the initialization distinguishes this stage from single-task fine-tuning.

Findings. The effect is consistent across all three merging methods: the brief fine-tune resolves the interference introduced by merging and recovers multi-task quality that the zero-shot merges never reach (Table 1).

5. Discussion and Conclusions

We set out to collapse a three-stage astronomical restoration pipeline – star removal, dead/hot-pixel restoration

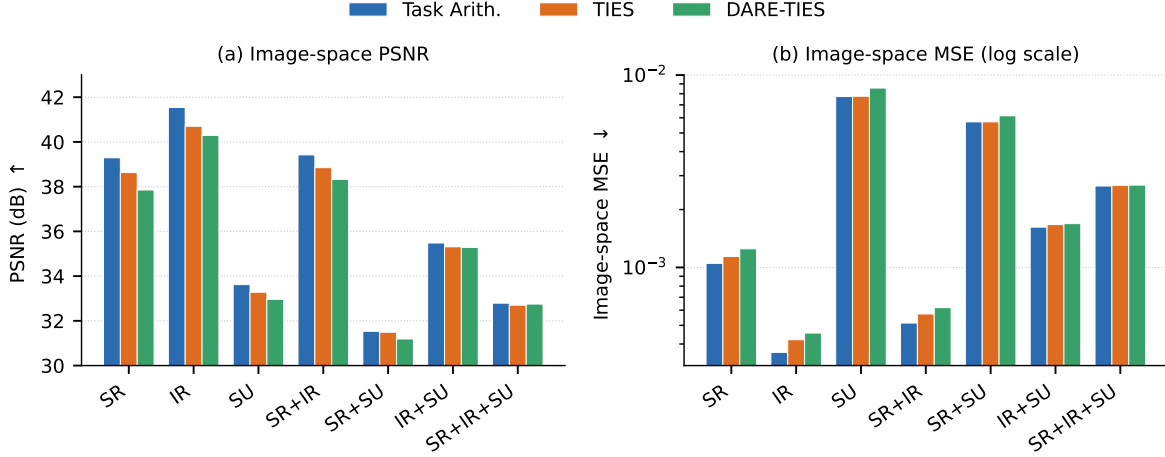


Figure 1. Per-folder comparison of the three merging methods on the image-space metrics of Tables ?? and ??: (a) PSNR (higher is better) and (b) MSE on a logarithmic scale (lower is better). The three methods track each other closely on most folders, with differences concentrated on the super-resolution-heavy sets.

and super-resolution – into a single set of weights by merging fully fine-tuned diffusion specialists that share a purpose-pretrained initialization θ_0 . Two findings stand out. First, evaluation is subtle: raw pooled MSE silently rewards suppressing the hardest task, and even after normalisation only a perceptual objective exposes the destructive interference between task vectors, which pixel-wise error cannot see (Sec. 4). Second, and more importantly, zero-shot merging is not enough on its own for these generative restoration tasks: a short, balanced post-merge fine-tuning stage removes the interference and lifts every merged model’s per-folder perceptual Z-scores from worse- to better-than-average (Table 1), recovering multi-task quality that none of the zero-shot merges reach.

[TODO: make this shorter]

Notably, Task Arithmetic, TIES and DARE-TIES converge to essentially the same quality once this stage is applied. In other words, the post-merge step is what makes weight-space merging viable for these generative restoration tasks in the first place, and the specific merging algorithm matters far less than whether this stage is applied at all. A practical consequence is that, for low-level diffusion-based restoration, one can favour the simplest merging rule and spend the compute budget on a brief joint fine-tune instead of on searching for an elaborate merging recipe. **[TODO: make this shorter]**

The study is limited by the small, manually-curated HLA set and the fixed three-task scope; whether the same post-merge recipe scales to more corruptions, larger archives, and more than three specialists – and how its cost compares to training a single multi-task model from scratch – is left to future work.

6. Statement on the use of AI

Three AI assistants were used in distinct, clearly scoped roles:

- *Claude* assisted with code generation, architectural design decisions, debugging during training and sampling, and drafting this report.
- *Gemini* was used only for code generation.
- *ChatGPT* was used only for editing and polishing the writing of this report.

All architectural choices, training runs, and quantitative results reported here were produced, checked, and validated by the authors.

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256 $\rightarrow$ 512 with self-attention (2 heads) at the deepest encoder stage, the bottleneck, and the first decoder stage.

TIES merging steps. For completeness, TIES-Merging combines task vectors $\{\tau_i\}$ in three steps: *trim* each τ_i to its top- $k\%$ largest-magnitude coordinates (zeroing the rest); *elect* a per-coordinate sign by summing the trimmed vectors and taking the sign of the aggregate; and *merge* by averaging, per coordinate, only those entries whose sign agrees with the elected sign. DARE-TIES additionally applies random per-coordinate dropout with rescaling to each τ_i before this procedure.

Appendix

Degradation hyperparameters. Star patches are typically 64×64 px and re-injected by linear blending onto starless backgrounds. Dead/hot pixels are detected from the residual to a 5×5 median filter with a 6.0σ threshold, σ estimated from the MAD scaled by 1.4826; hot pixels approach the clipping value (1.0) and dead pixels drop toward the local noise floor (0.0). Super-resolution pairs use Gaussian blur ($\sigma = 1.5$), $\times 4$ cubic down-sampling, and bicubic upsampling back to full resolution. The θ_0 pretext task applies 1–3 random operations from {Gaussian/Poisson-like noise, Gaussian blur, bicubic downsample–upsample, small local occlusions} with a 10% zeroed-condition dropout.

Training hyperparameters. All models use the v -prediction parameterization, a cosine schedule with $T = 200$ steps, EMA ($\beta = 0.999$), AdamW with learning rate 1×10^{-4} and weight decay 1×10^{-4} , and cosine annealing over 100 epochs. The U-Net widths are $64 \rightarrow 128 \rightarrow$